Does prefix validation select better predictions?

Exploratory external benchmark appendix | Retain the prior model

We tested whether adding a guarded position-regression candidate to the existing selector improves forecasting. The completed experiment does not support promotion: SoftVT predictions are unchanged, while the PhysTwin aggregate worsens. Earlier results and forecasts remain preserved.

Protocol and scope

All 140 SoftVTBench sequences and eight PhysTwin sequences had been examined previously. This is an exploratory algorithm ablation, not a fresh confirmation set or a transfer evaluation of the new coupled FEM solver. Initial mechanical geometry and known controller trajectories are provided; the experiment does not reconstruct these inputs from images.

The candidate uses fixed ridge regularization 0.1, existing FEM-anchor volume repair and the same static contact approximation. Its first 70% observed prefix fits the model, and the remaining observed prefix validates it against the archived selected configuration reconstructed from independent short fits. A switch requires strictly more than 2% lower validation error. Otherwise the archived forecast is retained exactly. The original selector also used this validation interval.

Complete-cohort prediction results

Method	SoftVT mm / complete	PhysTwin mm / complete
New guarded ridge switch	20.902 / 140/140	37.044 / 8/8
Prior selected model	20.902 / 140/140	33.019 / 8/8
Rigid model	20.513 / 140/140	66.788 / 8/8
Prefix-calibrated rigid	21.826 / 140/140	87.438 / 8/8
Raw support-space ridge	179.004 / 140/140	29.705 / 8/8
Official PhysTwin baseline	Not applicable	21.144 / 8/8

SoftVT values are coordinate RMSE; PhysTwin values are manual-track mean distance. Do not pool the two metrics. Coverage accompanies complete-case means. The raw support-space ridge differs from the guarded nodal candidate in interpolation and feasibility processing. This is not an official SoftVT policy benchmark.

Selection decisions and limitations

SoftVTBench: 0/140 switched. Against the prior selected model, 0 improve, 0 worsen and 140 tie across 140 complete matched cases. Mean reduction in error: 0.000 mm (positive favors the extension).

PhysTwin: 3/8 switched. Against the prior selected model, 1 improve, 2 worsen and 5 tie across 8 complete matched cases. Mean reduction in error: -4.025 mm (positive favors the extension).

Every switched case

Case	Prefix old/new mm	Future old/new mm
single_lift_sloth	14.298 / 11.531	37.116 / 47.450
single_lift_dinosor	10.972 / 9.939	18.119 / 13.816
single_push_sloth	36.833 / 36.095	67.627 / 93.799

Prefix values are observed-position coordinate RMS. Future values use the dataset metric stated on page 1. They cover different intervals and, for PhysTwin, different observation metrics; their absolute magnitudes are not directly comparable.

What this result establishes

An improvement on the observed validation interval did not reliably select a better future predictor. Geometry safeguards can preserve valid elements without producing accurate motion. These observations motivate better treatment of changing interactions; they do not establish that a particular replacement architecture or uncertainty model will solve the problem.

What remains unchanged

We retain the prior selected model and all previous benchmark results. No threshold, candidate, or per-case choice was changed after scoring. The new results are preserved as a negative ablation. Success in the separate controlled simulation study must be assessed on its own held-out cohort and cannot substitute for external benchmark gains.

Audit and reproducibility

All 148 forecasts were saved before evaluator access. Independent verification reconstructs ridge features, coefficients, short-prefix losses, selection decisions and new forecast metrics; archived comparator payloads are integrity-checked, not retrained. Frozen geometry/contact primitives are reused, so this is not an independent alternative physics solver. Raw datasets and prediction arrays remain local. Protocols, hashes, compact results and reproduction notes accompany this appendix.

Evidence: reports/prefix-ridge-switch-v13/{protocol.json, scores.json, summary.json, independent-audit.json, REPRODUCE.txt}. Source: scripts/prefix_ridge_switch_v13. The historical run enforces its original deadline; a later replay requires a separately declared protocol and matching local assets.